

Falsification of Autonomous Driving Policies using Fuzzing and Importance Sampling

John Greaney-Cheng
Stanford University
johngc@stanford.edu

Mete Gumusayak
Stanford University
mete1@stanford.edu

Justin Hall
Stanford University
jhall2025@stanford.edu

Annie Villalta
Stanford University
anniev18@stanford.edu

Abstract—Autonomous driving systems rely heavily on reinforcement learning (RL) policies trained through simulation and real-world data. However, even well-trained policies can fail under rare environmental conditions. In this project we investigate simulation-based falsification techniques for identifying failure scenarios in autonomous driving environments.

We evaluate two complementary approaches: fuzzing-based stress testing and importance sampling for failure probability estimation. Using the Highway-Env simulation framework, we introduce both action and environment disturbances into the merge track on Highway-Env to discover high-likelihood failure trajectories. Our experiments analyze the efficiency of different proposal distributions in generating failures and estimating failure probabilities.

Results demonstrate that carefully designed proposal distributions can accelerate the discovery of safety-critical scenarios compared to naive random sampling, given the agent is trained effectively using an applicable RL algorithm. These findings highlight the importance of targeted falsification methods for evaluating safety-critical autonomous systems.

I. INTRODUCTION

Autonomous vehicles must operate safely in complex and highly dynamic environments. Modern control policies for autonomous driving are often developed using reinforcement learning (RL), which enables agents to learn decision-making strategies through interaction with simulated environments. While RL policies can achieve strong performance in standard operating conditions, guaranteeing safety remains a major challenge. Even well-trained policies may fail under rare or unexpected situations that are unlikely to appear during training.

Traditional testing methods based on large numbers of random simulations are often inefficient at discovering these rare but safety-critical failure scenarios. As autonomous systems become more complex, identifying such failures through naive sampling becomes increasingly difficult. Simulation-based falsification techniques aim to address this challenge by intentionally searching for trajectories that violate safety properties, allowing developers to uncover weaknesses in control policies before deployment.

In this project, we investigate simulation-based falsification methods for reinforcement learning agents in autonomous driving environments. Our experiments are conducted using the Highway-Env simulation framework, focusing specifically on the highway merge scenario, which represents a common and safety-critical driving situation. The merge environment

introduces interactions between vehicles traveling at different speeds and lanes, making it a useful benchmark for evaluating autonomous driving policies.

To explore potential failure scenarios, we introduce several types of disturbances that perturb the agent’s behavior and environment conditions. These include random action perturbations, delayed actions that repeat previous control inputs, and variations in the initial state distribution of vehicles on the roadway. By systematically introducing these disturbances, we aim to identify high-likelihood failure trajectories and better understand the conditions under which collisions or unsafe behaviors occur.

The goal of this work is twofold: first, to discover failure scenarios through fuzzing-based stress testing, and second, to estimate the probability of such failures under nominal operating conditions using importance sampling techniques. Together, these approaches provide insight into both how failures occur and how likely they are, contributing to more robust evaluation methods for safety-critical autonomous systems.

II. PROBLEM DEFINITION

We model the autonomous driving system as a policy interacting with a stochastic simulation environment—the merge track on Highway-Env.

Our agent car observes the state of surrounding vehicles and selects discrete driving actions based on the scenario it is in. The actions include:

- `LANE_LEFT`: Move left one lane, keep speed constant
- `LANE_RIGHT`: Move right one lane, keep speed constant
- `IDLE`: Don’t change lanes, keep speed constant
- `FASTER`: Don’t change lanes, speed up
- `SLOWER`: Don’t change lanes, slow down

A failure episode occurs if our agent vehicle and another vehicle collide at any timestep in the simulation. Otherwise, we mark the episode as a success.

The objectives of this project are the following:

- Estimate the probability of failure under nominal conditions.
- Identify the most likely failure trajectories in the simulation environment.
- Evaluate the efficiency of different falsification methods.
- Utilize importance sampling techniques to attain an estimate of the probability of failure, or P_{fail} .

III. LITERATURE REVIEW

A. Highway Env and Reinforcement Learning

Chen et al. [4] study decision-making for autonomous vehicles in highway environments using deep reinforcement learning. The authors formulate the driving problem as a sequential decision task and apply a Proximal Policy Optimization (PPO)-based reinforcement learning algorithm to learn driving policies over a continuous action horizon. Their framework models vehicle kinematics and highway traffic interactions while optimizing for safe and efficient driving behavior. Experimental results demonstrate that the proposed approach achieves improved learning efficiency and stable driving performance in simulated highway scenarios, highlighting the potential of reinforcement learning for autonomous vehicle decision-making.

Healy et al. [6] study failure conditions for autonomous vehicles performing mandatory lane changes in a simulated highway environment. Using the Highway-Env framework, the authors evaluate the reliability of a Deep Q-Network (DQN) agent under sensor and control disturbances. In their experimental setup, observation noise is modeled with Gaussian perturbations applied to vehicle state measurements, while action disturbances are modeled using a probabilistic mechanism in which the agent follows its intended action with high probability and executes a random action otherwise. This formulation allows the study of how small stochastic deviations in perception and control can lead to rare but critical failure events such as collisions. The work demonstrates how techniques such as fuzzing and importance sampling can be used to efficiently identify and analyze these failure scenarios.

B. Adaptive Importance Sampling

Adaptive Importance Sampling (AIS) was introduced as a method to improve the efficiency of Monte Carlo integration by iteratively adjusting the proposal distribution to better match the target distribution. Oh and Berger [5] proposed one of the foundational AIS schemes, demonstrating how adaptive selection of sampling densities can significantly reduce variance compared to standard importance sampling. Their approach provided a systematic framework for updating the proposal based on previously observed samples, establishing the basis for many modern adaptive strategies in both Bayesian inference and reinforcement learning contexts.

IV. APPROACH

A. Simulation Environment

For our experiments, we utilized the Merge environment from the HighwayEnv simulator. The scenario is defined with a discrete 18-timestep horizon, where the agent's goal is to safely navigate interactions between fast-moving highway traffic and a slower merging vehicle. We specifically categorize observed failures into two distinct types: interactions with highway vehicles (highway failures) and interactions with the merging vehicle (merge failures). This environment is pictured in Figure 1.

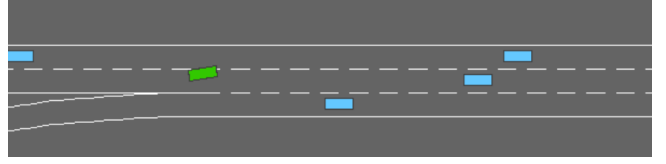


Fig. 1. Merge Track in Highway-Env. The green car represents our agent.

B. Fuzzing-Based Falsification

We investigate two falsification techniques: probabilistic fuzzing using random action and delayed action probabilities, and initial state fuzzing. These fuzzing techniques introduced controlled perturbations into the system to increase the likelihood of encountering failures.

Each proposal distribution is evaluated by running $m_{fuzz} = 20,000$ rollouts and measuring the likelihood of the most probable failure trajectories discovered.

1) *Probabilistic Fuzzing*: As mentioned above, we attempt two different probabilistic fuzzing techniques, deterministically selecting either the intended action from our agent's policy, or a perturbed action.

Delayed Action: With probability p_d , choose the previous timestep's intended action, essentially functioning as a "delayed action" as it occurs one timestep later than it should. The intended action is selected with probability $1 - p_d$. The nominal distribution for our policy introduces $p_d = 0.01$ to incorporate randomness to our otherwise deterministic policy. The selected delay are probabilities for fuzzing were the following: $[0.02, 0.03, 0.05, 0.075, 0.1, 0.2, 0.3, 0.4, 0.5]$. We aimed to include a mix of values that were close to the nominal and far from the nominal to analyze a trade-off in predicting the most likely failure while increasing the number of observed failures.

Random Action: With probability p_r , choose a completely random action at uniform. With probability $1 - p_r$, choose the intended action. The nominal in this case is $p_r = 0.01$, and the selected random action probabilities were the following: $[0.02, 0.03, 0.05, 0.075, 0.1, 0.2, 0.3, 0.4, 0.5, 1]$. $p_r = 1$ was included to have a fully uniformly random fuzzing distribution for experimentation.

2) *Initial Condition Fuzzing*: We conduct Initial Condition (IC) fuzzing to test the environment's starting states. We define a joint Importance Sampling proposal over three initial state variables for each car besides ego and merger:

- **Lane Assignment**: Sampled from a Bernoulli distribution with parameter q_{lane} (nominal $q = 0.5$).
- **Position Offset**: Sampled from a $\text{Beta}(\alpha, \alpha)$ distribution scaled to $[-5, 5]$ meters.
- **Speed Offset**: Sampled from a $\text{Beta}(\alpha, \alpha)$ distribution scaled to $[-1, 1]$ m/s.

Under nominal conditions, the continuous offsets are drawn from a uniform distribution (equivalent to $\alpha = 1$). Our fuzzing proposal sets $\alpha < 1$, creating a U-shaped probability density. This heavily biases the initialization toward extreme spatial and velocity configurations, placing vehicles in more extreme starting states that could lead to more failures. Initial Condition

fuzzing is exclusively on the highway vehicles because in the nominal environment, the merging and ego vehicles' initial position and speed are deterministic

C. Generating "Ground Truths"

To get a baseline estimate for P_{fail} to use for each of our fuzzing techniques, we sampled $m_{direct} = 100,000$ rollouts for each technique with the nominal distribution. Our agent ultimately was trained using Proximal Policy Optimization and was found to be roughly $P_{fail} = 0.02$ generally before applying any stochasticity to the policy, which we deemed adequate for experimentation for fuzzing and importance sampling, but not too low to the point where it would take many days to run experiments. We argue that these can be assumed as ground truths for the paper due to the tight 95% confidence intervals that we found, specifically being $[0.02002, 0.02180]$ for delayed action ($p_{fail} = 0.02091$) and $[0.02900, 0.03112]$ for random action ($p_{fail} = 0.03006$) after m_{direct} episodes.

D. Importance Sampling

Importance sampling estimates the probability of failure under the nominal system distribution while sampling from a different proposal distribution that produces failures more frequently.

For each sampled trajectory τ , we compute an importance weight:

$$w(\tau) = \frac{p(\tau)}{q(\tau)}$$

where $p(\tau)$ is the nominal distribution and $q(\tau)$ is the proposal distribution used for sampling.

This allows unbiased estimation of the failure probability even when sampling from a biased distribution designed to expose failures.

V. RESULTS

A. Direct Sampling

Figures 2 and 3 highlight the convergence of P_{fail} with the nominal values of $p_d = p_r = 0.01$ over $m_{direct} = 100,000$ samples. It should be noted that random action fuzzing is generally more aggressive than delayed action fuzzing, as delayed action does still pick an action from the agent's intended policy one timestep prior, but random actions is completely up to chance, with a 20% chance of taking any action in the space. Delayed action converges to roughly $p_{fail} = 0.02091$, whereas random action converges to $p_{fail} = 0.03006$, which is roughly 43.8% higher.

Figure 4 shows where failures occur most often under the nominal distribution for the initial condition fuzzing. The predominantly red areas indicate that most failures happen when vehicle 0 and vehicle 1 are both in the left lane. This situation could be studied further to understand why it leads to crashes for our agent.

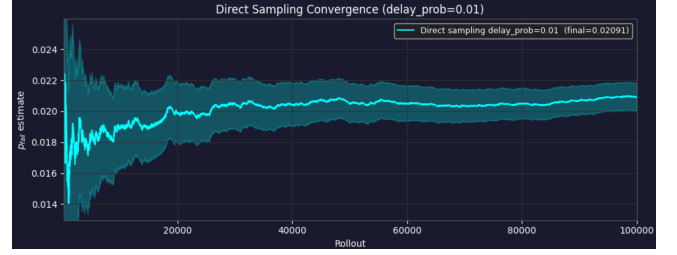


Fig. 2. Delayed Action Direct Sampling

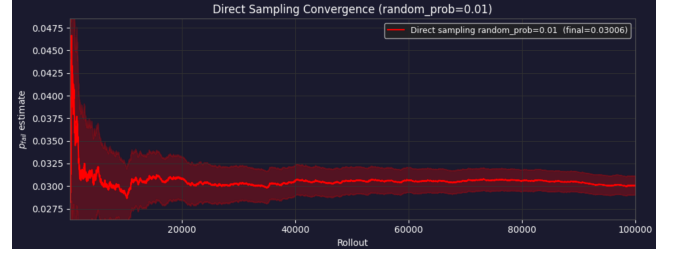


Fig. 3. Random Action Direct Sampling

B. Fuzzing and Most Likely Failure

The calculated failure log-likelihood distributions for random action and delayed action were quite distinct. As shown in Tables I and II, the number of failures tended generally grew linearly with an increase in p_r , but tended to be more unpredictable with an increase in p_d , though generally trending upwards.

delay_prob	failures	mlf_logp
0.020	388/20000	-0.1005033585
0.030	450/20000	-0.1005033585
0.050	414/20000	-0.1005033585
0.075	456/20000	-0.1005033585
0.100	418/20000	-0.1005033585
0.200	426/20000	-0.1005033585
0.300	503/20000	-0.1005033585
0.400	537/20000	-4.6956232087
0.500	592/20000	-4.6956232087

TABLE I

FAILURE STATISTICS FOR VARYING DELAY PROBABILITIES.

random_prob	failures	mlf_logp
0.020	754/20000	-0.1105536944
0.030	1007/20000	-0.1105536944
0.050	1441/20000	-0.1105536944
0.075	1876/20000	-0.1105536944
0.100	2445/20000	-0.1105536944
0.200	4314/20000	-0.1105536944
0.300	6121/20000	-0.1105536944
0.400	7576/20000	-6.2749101135
0.500	9119/20000	-12.4493168686
1.000	17109/20000	-24.8584323937

TABLE II

FAILURE STATISTICS FOR VARYING RANDOM PROBABILITIES.

Interestingly, for all three of delayed action, random action, and initial state fuzzing, the most likely failure probability, (mlf_logp), converged to the same type of trajectory: an $z =$

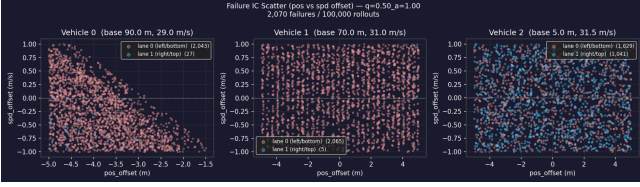


Fig. 4. Nominal Initial Condition Failure Distribution

11 timestep collision with no perturbed actions (11 successful timesteps then a failure on step 12). This can be proven by the fact that our nominal distribution was 0.01, and

$$p_r = p_d$$

$$p_{\text{success}} = 1 - p_r$$

$$\ln(p_{\text{success}})z = \ln(0.99)11 = -0.110553694$$

Which is identical to `mlf_logp` for random action. Notice that if z was 10 instead of 11 for delayed action, the corresponding `mlf_logp` for delayed action would be achieved. This is because delayed action skips the first timestep, as there is no previous action. Lastly, because the initial state calculation does not use perturbed actions, all three fuzzing methods converged on the same most likely failure. The actions of the agent in this trajectory can be seen on Table III; the video corresponding to the most likely failure can be found in the Appendix. The most likely failure is our agent driving forward for the entire duration of the trajectory before being cut-off by an external agent without having time to turn/slow down.

Time Step t	Action
1	SLOWER
2	SLOWER
3	SLOWER
4	SLOWER
5	IDLE
6	IDLE
7	IDLE
8	IDLE
9	IDLE
10	IDLE
11	IDLE

TABLE III

ACTION SEQUENCE FOR THE MOST-LIKELY FAILURE TRAJECTORY.

Furthermore, the red Figure 5 represents the failure log-likelihood distribution for *all* all random action proposals pooled. The light-blue figure 6 represents the failure log-likelihood for all delayed action proposals pooled. The last figure shows an example of an individual proposal distribution for delayed action, being $p_d = 0.02$. The random action distribution generally covers actions which are much more unlikely, largely due to more proposals that are extremely dissimilar to the nominal in regard to Kullback-Leibler (KL) divergence.

C. Importance Sampling (IS)

Across a delayed action proposal sweep, the IS estimates for nominal failure probability remain close to the direct-sampling

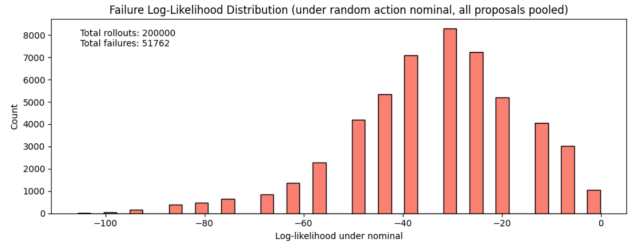


Fig. 5. Random Action Failure Distribution Across All Proposals

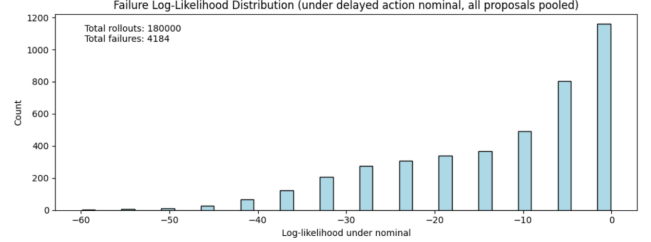


Fig. 6. Delayed Action Failure Distribution Across All Proposals

baseline ($\approx 2\%$) for moderate proposal strengths (e.g., delay probabilities between 0.01 and 0.2), while overly aggressive proposals (e.g., delay probability 0.5) lead to weight degeneracy and unstable estimates, illustrating the trade-off between inducing failures and maintaining an effective sampling distribution. Moreover, this generally remained consistent across the random action proposal sweep as well, with exception to the completely random uniform fuzzing distribution, which predicted a value of $P_{\text{fail}} = 0$. Tables IV and V show the optimal proposal values we found using importance sampling for delay and random action.

delay_prob	p_fail_is	p_fail_error
0.020	0.01932	0.00159
0.030	0.02212	0.00121
0.050	0.02121	0.00030
0.075	0.02137	0.00046
0.100	0.01978	0.00113
0.200	0.02076	0.00015
0.300	0.02274	0.00183
0.400	0.00202	0.01889
0.500	0.00194	0.01897

TABLE IV

IMPORTANCE SAMPLING ESTIMATES FOR DELAY PROBABILITIES (NOMINAL = 0.02091)

D. Adaptive Importance Sampling

Adaptive importance sampling was employed to estimate the merge-task failure probability under delayed actions. A Beta proposal over the per-step delay probability was iteratively refined via a cross-entropy update: each iteration collected 20 rollouts under the current proposal, weighted failures by the likelihood ratio between the sampled delay trace and the nominal 0.01 delay policy, and updated the proposal toward the elite failure samples while preventing collapse. After 1,000 iterations (20,000 rollouts), the estimator

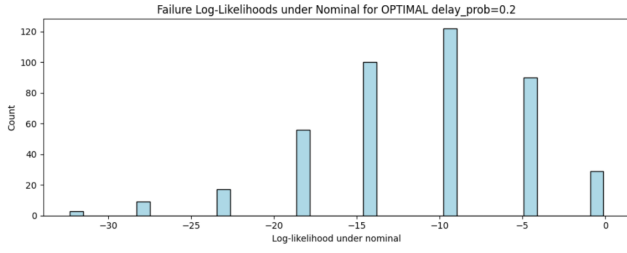


Fig. 7. Delayed Action Failure Distribution for $p_d = 0.2$.

random_prob	p_fail_is	p_fail_error
0.020	0.02870	0.00136
0.030	0.03059	0.00053
0.050	0.02967	0.00039
0.075	0.02783	0.00223
0.100	0.03190	0.00184
0.200	0.02717	0.00289
0.300	0.02787	0.00219
0.400	0.01166	0.01840
0.500	0.00278	0.02728
1.000	0.00000	0.03006

TABLE V

IMPORTANCE SAMPLING ESTIMATES FOR RANDOM PROBABILITIES (NOMINAL = 0.03006).

converged to a failure probability of approximately 0.0250 with a 95 percent confidence interval of $[0.0215, 0.0286]$, and the Beta mean stabilized near 0.068, indicating that the learned proposal concentrates probability mass on mildly more aggressive delay schedules without departing from realistic behavior.

E. Proposal Optimization and Convergence

1) *Action Fuzzing: Delayed and Random Action:* As shown via the tables in Section V-B, we found the `delay_prob` = 0.2 and the `random_prob` = 0.05 to minimize the error between our nominal estimate via direct sampling over $m_{direct} = 100,000$ samples. Figures 8 and 10 show the convergence of importance sampling against direct sampling (the orange chart). In both cases of importance sampling, the distribution converges to the "ground truth", but has a much larger confidence interval than direct sampling, especially in the case of the delayed action importance sampling, which had a larger optimal proposal probability. These figures just show the first 20,000 episodes, but it should be noted that the orange direct sampling chart was run over 100,000 episodes.

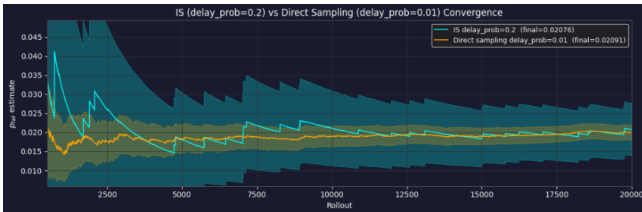


Fig. 8. Delayed Action Convergence for $p_d = 0.2$.

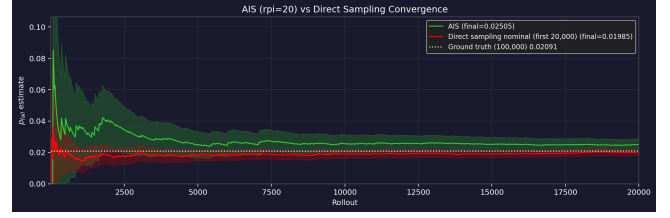


Fig. 9. Convergence of the AIS failure-probability estimate over 1,000 iterations, with a shaded 95 percent CI showing the estimate stabilizing near 0.0250.

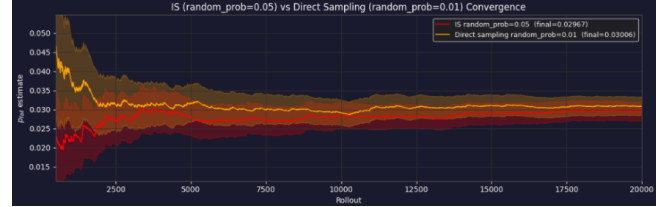


Fig. 10. Random Action Convergence for $p_r = 0.05$.

2) *Environment Fuzzing: Initial State Fuzzing:* To determine the optimal parameters for our initial state proposal, we swept across the grid of proposal parameters mentioned in the approach section ($q_{lane} \times \alpha$) for 20,000 rollouts each. We then compared the estimated probability of failure ($\hat{P}_{fail}$) against the Direct Sampling ground truth baseline with 20762 failures over 1,000,000 nominal rollouts ($P_{fail} \approx 0.02076$).

Table VI shows the parameter configuration $q_{lane} = 0.10$ and $\alpha = 1.00$ was identified as the optimal proposal. It produced an Importance Sampling $\hat{P}_{fail}$ estimate of 0.02092 which was closest to the 0.02072 ground truth and generated 1328 failures within 20,000 rollouts. Figure 11 illustrates the convergence of this IS estimate compared to the 1M Direct Sampling baseline.

Configuration	Failures	Highway	Merger	IS P_{fail}
$q = 0.30, \alpha = 0.30$ (OPT)	1,426	1,426	0	0.02481
$q = 0.30, \alpha = 1.00$	795	795	0	0.02131
$q = 0.50, \alpha = 1.00$ (NOM)	441	441	0	0.02205
$q = 0.10, \alpha = 0.30$	2,289	2,289	0	0.01839
$q = 0.10, \alpha = 0.50$	1,882	1,882	0	0.01962

TABLE VI

THE OPTIMAL AND SECOND-BEST IMPORTANCE SAMPLING ESTIMATES, AND THE TWO CONFIGURATIONS WITH THE LARGEST NUMBER OF FAILURES OVER 20,000 ROLLOUTS

VI. DISCUSSION

Our experiments show that targeted proposal distributions can recreate likely failure trajectories and provide accurate estimates for the probability of failure for our agent on the merge system. Fuzzing methods quickly expose diverse failure scenarios, while importance sampling provides a statistically grounded method for estimating failure probabilities under the nominal system distribution.

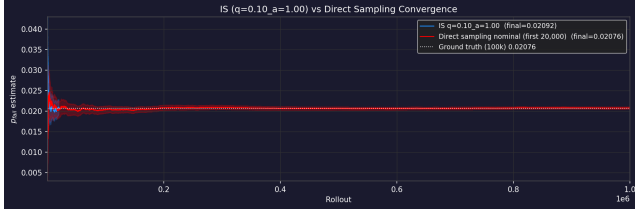


Fig. 11. Convergence of $\hat{P}_{fail}$ using Importance Sampling with the optimal proposal (20k rollouts) versus the full Direct Sampling ground truth baseline (100k rollouts).

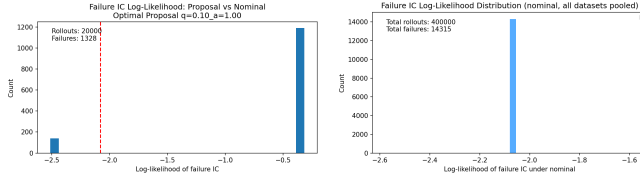


Fig. 12. Initial Condition Fuzzing Likelihoods. Left: The optimal proposal ($q = 0.10, \alpha = 1.00$) peaks at -2.4 and -0.3 . Right: The pooled nominal distribution exactly at $\log(2^{-3})$.

However, several limitations remain. The largest limitation of this entire paper is that ultimately, while the importance sampling and adaptive importance sampling algorithms were able to converge to the correct P_{fail} given our assumption that the direct sampling 100,000 episode run is the ground truth, they were not able to converge faster than direct sampling. This is an agent training issue, as without any perturbations to our policy after training, our agent completed merge roughly 98% of the time. This leaves a 2% failure rate.

$P_{fail} = 0.02$ makes a failure event uncommon, but certainly not rare. A much lower value, $P_{fail} < 0.001$ would be required to meet this, and this is where importance sampling would see drastic improvements when compared to direct sampling.

With more time, we likely would have trained a better agent to achieve a lower failure rate, which is a lesson learned, as we originally thought 2% was a balance between being high enough to do falsification effectively, but low enough to see benefits with importance sampling. However, this balance proved to be more nuanced (and much lower) than we thought it was.

Other limitations include both the sensitivity to the choice of proposal distribution and computational cost for large-scale simulations, as some of our experiments took upwards of a dozen hours to run.

Furthermore, our initial goal was to conduct this falsification analysis in a racetrack environment. However, we pivoted to the highway merge environment because it was difficult to train a stable, competent baseline reinforcement learning agent on the racetrack (the lowest P_{fail} value we were ever able to achieve via training was 0.32).

VII. CONCLUSION

This project demonstrates the effectiveness of simulation-based falsification methods for evaluating autonomous driving

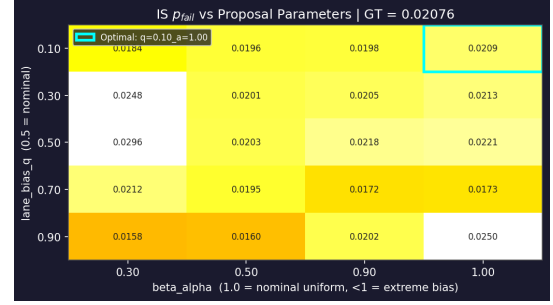


Fig. 13. Importance Sampling $\hat{P}_{fail}$ versus IC Proposal Parameters ($q_{lane} \times \alpha$). The configuration $q = 0.10, \alpha = 1.00$ best matches the Direct Sampling ground truth.

policies. By combining fuzzing and importance sampling, we identify failure trajectories and estimate failure probabilities more efficiently than naive simulation approaches.

Such techniques are critical for improving the safety and reliability of autonomous systems before deployment in real-world environments.

A. Future Work

Several extensions could further improve the robustness of falsification analysis:

- Adaptive importance sampling methods
- Cross-entropy optimization for proposal distributions
- Bayesian failure probability estimation
- Application to more complex autonomous driving environments

Our falsification methods successfully induced failures, but fuzzing resulted only in "highway" failures, missing "merge" failures. For future work, we recommend designing multiple failure types to be more evenly distributed, for a more balanced analysis of differing failures.

These techniques could enable more efficient discovery of rare safety-critical failures.

VIII. CONTRIBUTIONS

John, Justin and Annie all took this class for 4 units, while Mete took this class for 3 units.

- **John:** Developed and implemented the Delay Action and Random Action fuzzing methodologies used to perturb the agent's control policy during simulation. Designed and implemented the experimental framework for evaluating these disturbance distributions across simulation rollouts. Implemented importance sampling algorithms for both Delay Action and Random Action disturbances, along with an adaptive importance sampling method for the Delay Action distribution to improve failure discovery efficiency. Identified and analyzed Most Likely Failure (MLF) trajectories and generated corresponding failure visualization videos for Delay Action and Random Action disturbance.
- **Mete:** Developed and implemented the Initial Condition (IC) fuzzing methodology. Trained the Proximal Policy

Optimization (PPO) merge agent used throughout the experiments. Executed the large-scale 1,000,000-episode Direct Sampling baseline for ground truth estimate. Worked on initial environment setup, testing, and agent training attempts for the racetrack environment before the project pivoted to the highway merge scenario.

- **Justin:** Initially attempted to train agents on the racetrack environment using Deep Q Networks overnight before realizing that racetrack was difficult to discretize. After pivoting to merge, my predominant role was creating the overnight scripts to run the validation experiments. Specifically for the action fuzzing experiments, with the help of John, I tried additional values to test for both random and delay action. I created all of the plots related to those action fuzzing distributions. I also ran the 100k direct sampling experiments for merge for delayed action and random action. For the paper itself, I focused on / wrote a significant portion of the problem definition, approach, results, and discussion sections.
- **Annie:** Ran Adaptive Importance Sampling across three different trials to estimate the delayed-action failure probability in the HighwayEnv merge task (Included designing the Beta proposal + cross-entropy updates, running three distinct trials over 20k rollouts each). Generated data/plots that rely on the merge PPO to show AIS convergence vs direct sampling. Initially trained agents using PPO on Highway, DQN with Highway, and racetrack with PPO. For the paper, I focused on writing the literature review and AIS sections.

REFERENCES

- [1] E. Leurent, “HighwayEnv: An Environment for Autonomous Driving Simulation,” 2018.
- [2] J. Schulman et al., “Proximal Policy Optimization Algorithms,” 2017.
- [3] M. Kochenderfer, T. Wheeler, and K. Wray, *Algorithms for Decision Making*. MIT Press, 2022.
- [4] H. Chen, X. Tang, and T. Liu, “Decision-making for autonomous vehicles on highway: Deep reinforcement learning with continuous action horizon,” arXiv preprint arXiv:2008.11852, 2020.
- [5] M. S. Oh and J. O. Berger, “Adaptive importance sampling in Monte Carlo integration,” *Journal of Statistical Computation and Simulation*, vol. 41, no. 3–4, pp. 143–168, 1992.
- [6] E. Leurent, “An Environment for Autonomous Driving Decision-Making,” paper 024.

APPENDIX

A. Video to Most Likely Failure

View Most Likely Failure Video